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A hybrid deep neural net learning model for predicting Coronary Heart Disease using Randomized Search Cross-Validation Optimization

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ABSTRACT

Coronary Heart Disease (CHD) is a life-threatening public health problem. Many chronic CHDs and health risks can be avoided, reversed, and reduced with proper risk assessment. Medical professionals find it challenging to anticipate heart attacks and heart failures since it is a complex process requiring knowledge, experience, and medical resource facilitation. Although healthcare is generally information-savvy, not all available data are analysed to find hidden patterns and make informed and timely decisions since heart disease prediction relies heavily on clinical data processing. This study proposes a hybrid deep neural net learning model for predicting CHD using the BRFSS-2015 Dataset. The best features subset is chosen based on the co-relation score and dataset classes are balanced using the cluster-abundant data class approach. Bi-direction Long Short-Term Memory (BiLSTM) and Gated Recurrent Unit (GRU) hyper-parameter tuning is accomplished using Randomized Search Cross-Validation Optimization (RSCV). In comparison to GRU, LSTM, and BiLSTM-GRU, this suggested model obtains a classification accuracy of 98.28% which outperforms existing models.

1. Introduction

According to the World Health Organization (WHO) [1], Cardiovascular and Cerebrovascular Disorders (CCVD) are the leading cause of human mortality, almost 18 million people worldwide pass away from CCVD every year. Conditions responsible for such high mortality rates are high blood pressure, obesity, high cholesterol, and hyperglycemia [2]. The leading cause of death from CCVD and CHD accounts for four-fifths of all deaths [3]. The prevalence of CHD is still increasing on a global scale. According to projections made by WHO in 1999, CHD would account for the majority of mortality cases worldwide by 2020 and will be the primary cause of death in developing economies. In 2019, data collected by WHO stated that cardiovascular disorders were responsible for 38% of the 17 million premature deaths (before the age of 70) caused by non-communicable diseases. According to data gathered by the British Heart Foundation [4], around 9.8 million men and 9.2 million women died from heart and circulation disorders worldwide in 2019. More than 23 million deaths per year from heart and circulation disorders are expected worldwide by 2030, and more than 34 million by 2060. Around the world, 550 million people are thought to be affected by heart and circulation disorders. The prevalence of CHD is rising daily, as a result of altering lifestyles, an ageing and expanding population, and better survival rates from heart attacks and strokes, and it will continue to increase if these trends continue. making it a serious public health issue that affects a nation's social and economic development. In order to reduce the risk of the high-risk categories, it is crucial to increase CHD prevention and treatment by undertaking an acceptable scientific risk assessment.

The major cause of Coronary Heart Disease (CHD) also known as Coronary Artery Disease (CAD) is the blockage of arteries that supply blood to the heart muscles [5]. Arteries get blocked as a result of calcium and cholesterol plaque deposition. This disorder is known medically as atherosclerosis. A thin strip of cells called the endothelium ensures that blood flows through arteries without interruption. In the case when the endothelium deteriorates, plaque builds up in the artery wall. Plaque buildup, which weakens blood cells, impedes the blood flow to the heart. If CHD is not treated promptly, symptoms may worsen as it progresses. Plaque builds up over time and becomes harder. If the plaque ruptures, platelets may stick together to form blood clots in the arteries. It might completely stop the blood flow, which would cause a cardiac arrest [6]. Angiography is frequently used to diagnose CHD [7]. This treatment involves invasiveness and has several drawbacks such as infection, mild reaction to the contrast agent, are some minor complications whereas heart stoke, kidney damage, internal bleeding and anaphylaxis are some serious concerns [8]. In order to diagnose CHD using clinical data, non-invasive methods are necessary.

The research discussed in this article focuses on applying machine and deep learning approaches to predict cardiac problems [9]. Over

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the past decade, many researchers are working on heart disease classification to develop highly efficient machine learning and deep learning models. In recent years, the application of artificial intelligence and machine learning algorithms has grown in popularity due to the increased accuracy and efficiency of prediction [10,11]. The significance of research in this field is in the ability to create and choose models with the maximum accuracy and efficiency [12]. Heart disease classification is a task in which we use a preexisting heart disease dataset and train a machine learning model on it and try to classify the patients as actually suffering from heart disease or not. A heart disease dataset contains records of a patient's medical examination, their general details like age, gender, and daily habits, and the target column indicates whether or not the patient suffers from heart disease.

Most of the machine learning models developed for Heart Disease classification uses UCI heart disease dataset [13]. UCI heart disease dataset is refined but very small in size. Machine Learning algorithms requires large amount of data for training and testing phases. Also preferred is a refined dataset which does not have missing values and includes all the possible combinations of data elements. Preparing a refined dataset is an expensive task but fortunately there is availability of few datasets which have large and refined collection of data related to Heart issues. One such dataset is Behavioural Risk Factor Surveillance System (BRFSS) 2015 heart disease dataset [14] which is a state-based telephone survey. The Centers for Disease Control and Prevention (CDC) United States is the coordinating agency for this survey. Using telephone survey techniques, self-reported data on chronic diseases and health risk behaviours is gathered year-round.

After successful pre-processing of the dataset and extracting the suitable features, we use this dataset for training a machine learning model, which then attempts to categorize a random individual as a heart patient or not. Based on the dataset, the article utilized a deep learning approach to classify heart disease. The deep learning approach used in the paper is Bi-direction Long Short-Term Memory (BiLSTM) and Gated Recurrent Unit (GRU) which are tuned by randomized search cross-validation based optimization technique. Later, we compare our model's performance with some existing approaches and individual models in order to generate the reliability and validity of the proposed model.

1.1. Our contribution

BRFSS is a highly imbalanced dataset, which requires data refinement and pre-processing. We first perform data balancing by using cluster abundant data class method. Later we extracted the suitable features from the overall feature set using a correlation-based feature selection technique. Then we perform hyper-parameter tuning using the Randomized Search Cross-validation method. We then proceed with the classification process using LSTM, BiLSTM, and GRU deep neural nets. Throughout this process, we focus on our main objective of increasing classification accuracy using best-in-class methods.

1.2. Article organization

In Section 2, the paper discusses the related work in the field of heart disease prediction. Section 3 introduces the methodology that is used in this research work along with the dataset description and definitions of the used techniques. In Section 4, the proposed model is discussed with a pictorial representation of the steps followed for classification. Section 5 discusses the results obtained based on the evaluation metrics and validation curve. Lastly, Section 6 provides the concluding remarks and future perspectives in the field.

2. Related work

The section discusses some literature related to the paper, and the work done by authors, and researchers in the field of health informatics, especially coronary heart disease classification. Alotaibi et al. [15] used Rapid Miner and the proposed combination of five different machine learning algorithms namely; Decision Trees (DT), Naive Bayes (NB), Random Forest (RF), Logistic Regression (LR), and Support Vector Machine (SVM). The ensemble learning model showed significant improvement and increased accuracy in comparison to individual models. The authors used UCI Heart Disease Dataset and increased its size using augmentation techniques.

Chen et al. [11] used coronary computed tomography angiography images of patients and extracted bifurcation features from images using the morphometric method. For assessing the performance using the measured features, Machine Learning based classifier are used and polynomial SVM is identified with the highest accuracy.

Bharti et al. [16] presented three methods for predicting cardiac disease (without FS and outlier detection, with FS and no outlier detection, and with FS and outlier detection). According to them, FS along with outlier identification is superior to the other two approaches. The UCI heart dataset was trained with LR, KNN, SVM, RF, and DT classifiers, and the least absolute shrinkage and selection operation (LASSO) technique was used to pick the most important characteristics for heart illness. While deep learning achieved 94.2% accuracy, the KNN approach only achieved 84.8% accuracy.

Ghosh et al. [17] worked on UCI Dataset and employed RELIEF [18] and LASSO [19] feature selection mechanisms with classification. With a selection of 13 essential features from the dataset, they achieve the highest accuracy of 92.65% using Random Forest Bagging Method algorithm. With a selection of 11 features using the LASSO method they achieve the highest accuracy of 97.85% using the Gradient Boosting Method algorithm, and with a selection of 10 features using the RELIEF method they achieve the highest accuracy of 99.05% using RFBM algorithm.

Nitant et al. [20] have suggested a decision tree-based artificial neural network model for forecasting cardiac disease. In order to extract the key features from the input in each layer and utilize them as the input for the subsequent layer of the ANN model, the authors used the decision tree as the activation function in the layers of the ANN.

By the integration of heart datasets; Cleveland, Hungary, Long Beach, and Swiss, to understand heart disease, Kanagarathinam et al. [21] produced a dataset titled "Sathvi" consisting of 531 instances, 12 attributes, and no missing values. For the purpose of predicting cardiovascular problems, they applied Pearson's correlation coefficient approach. In this study, the CatBoost algorithm was used for classification and obtained an accuracy of 87.85%.

Gupta et al. [22] employed the Cleveland heart dataset to identify high-risk individuals with heart disease. The model learned several algorithms and standardized the data to the conventional scales. Using Logistic Regression, it achieved the greatest accuracy of 92.30%. The KNN classifier was also tweaked with k values ranging from 2 to 20; at $k = 14$, the classification accuracy was 90.11%.

Saboor et al. [23] predicted heart disease using the UCI heart disease dataset and standardized heart dataset features for optimal prediction results, then using the GridSearchCV technique to fine-tune the hyperparameters of machine learning classifiers. An accuracy of 96.72% was attained using a support vector machine (SVM) classifier with a sigmoid kernel and a complexity value of $C = 0.5$.

Using exploratory analysis of features from a dataset pertaining to heart disease, Chang et al. [24] created a python-based model of the Random Forest technique. A correlation matrix plot was used to assess the significance of the features in the heart dataset, and an accuracy of 82.18% was attained using a random forest classifier.

Reddy et al. [25] combined the Cleveland and Statlog cardiac datasets and extracted the finest core features using the CFS method

(Correlation-based Feature Selection). Second, utilizing three individual and three ensemble classifiers on datasets yielded the appropriate hyperparameters which depict the best prediction outputs. The author claims an accuracy of 97.91% with the Random Forest ensemble classifier.

Ozcan et al. [26] used an extensive dataset including data from five heart disease datasets of 1190 individual patients. To understand the connections between input and output data, the Classification and Regression Tree (CART) algorithm had been used to make predictions about cardiac disease. The author ranked the selected features based on their importance and achieved an accuracy of 87%.

Ogundepo et al. [27] used two datasets, the Cleveland dataset was considered for classification and the Statlog dataset was considered to validate the model. The authors performed an in-depth exploratory analysis of the Cleveland data using the Chi-square test of independence and followed by training ten classification models for prediction. The author stated support vector machine provided the most accurate predictions with a classification accuracy of 85% and validation accuracy of 87.04% with the Statlog dataset.

Using the BRFSS 2015 heart disease dataset, Fernando et al. [28] assessed the performance of various supervised classification models in terms of accuracy. These models included Naive Bayes, LightGBM, Decision Trees, Random Forest, XGBoost, K Nearest Neighbours, and ADABoost. Smoteen and SmoteTomek outperformed the other sample methods and were used by the authors to address the class imbalance problem in the dataset, achieving 97.10% classification accuracy with random forest model.

Using data from the 2015 BRFSS survey of US citizens, Das et al. [29] developed and compared six machine learning models for predicting cardiovascular disease. These models included Xgboost, Bagging, Random Forest, Decision Tree, K-Nearest Neighbour, and Naive Bayes. The six machine learning models were compared in terms of their accuracy, sensitivity, F1-score, and area under the curve (AUC). According to the authors, the Xgboost model produced the most optimal outcomes, with an accuracy rate of 91.30%.

Vardhan Shorewala [30] employed ensemble modelling strategies like bagging, boosting, and stacking to enhance coronary heart disease prediction accuracy. The 'Cardiovascular Disease Dataset,' which includes 70,000 records of patient data for coronary heart disease [31], is used to test the modelling approaches. The author concluded that the average accuracy for bagging was 73.82%, for boosting 73.89%, and for the stacked ensemble approach it was 75.1%.

A hybrid method, combining evolutionary algorithms and support vector machines for classification, was proposed by Sandhu et al. [32] To estimate the mortality rate of heart failure patients, the scientists analysed medical records from 299 people in the New York Health Associations' UCI cardiovascular disease categories III and IV patients. With their proposed SVM-GA model, the authors achieved an accuracy of 91.49%.

Bhatt et al. [33] applied k-modes clustering method over 70,000 patient records to predict heart disease. The classifiers were all hyper-tuned with GridSearchCV. Using multilayer perceptron with cross-validation, the authors improved classification accuracy to 87.28%.

Sridharan Kannan [34] developed an automated clinical decision support system for predicting cardiovascular disease utilizing linear support vector regression and stacked linear swarm optimization, achieving 98% classification accuracy.

Mondal et al. [35] Mondal et al. proposed a model to predict CHD using dataset consisting of 4240 instances. The best-performing boosting models (GB, HGB, XGB, CB, and LGBM) were taken into account for the final prediction, and a total of 13 ML models were deployed using 10-fold cross-validation. The accuracy of the predictions was 92.26%.

The research conducted by Nagavelli et al. [36] provides a comprehensive overview of available options for detecting cardiac disease using machine learning. With the goal of improving detecting heart disease, researchers used XGBoost to compare different decision tree classification methods. XGBoost achieves a 95.9% accuracy.

Dileep et al. [37] tested deep learning techniques on the UCI heart disease dataset and the real-time dataset. The authors suggested cluster-based bi-directional long-short-term memory (C-BiLSTM), which uses the K-Means clustering approach to remove duplicate data. According to the authors, C-BiLSTM is the best classification model, with 94.78% accuracy on the UCI dataset and 92.84% accuracy on the real-time dataset.

To predict CHD in its early phases, Chandrasekhar et al. [38] used GridSearchCV with five-fold cross-validation for hyperparameter optimization with six ensembled ML algorithms. The soft voting classifier predicted CHD with 93.44% accuracy on the Cleveland dataset and 95% accuracy on the IEEE Dataport dataset [39].

Kwakye et al. [40] proposed a number of machine-learning classification techniques to predict CHD in patients. The authors used the Kaggle Framingham dataset. SMOTE [41] was used to handle unbalanced classes and feature selection approaches are utilized. The authors measured accuracies on balanced and unbalanced datasets, and found that LR outperformed on imbalanced data while RF outperformed on balanced data.

Some research papers [42–47] predicted coronary heart disease on the different dataset, utilized different methodologies for data preprocessing and feature selection and achieved higher classification accuracy using random forest model. Also, some authors [48–51] have surveyed various machine learning and hybrid techniques for heart disease prediction.

In order to conclude the literature, there are certain limitations in the work of previous authors, some authors analysed a small amount of individual data for classification, and the models implemented with small data may not validate the model for bulk data. The state-of-art representation of the authors in the previous years is depicted in Table 1. The authors mainly used machine learning approaches for classification, while deep learning approaches may yield better results. The evaluation metrics in most of the literature are accuracy, sensitivity, specificity, and f1 score but some validation curve might be helpful for better understanding.

2.1. Motivation

As discussed in the previous Section 2 related work in the field of coronary heart disease prediction, and certain limitations. The paper motivates from the limitations and aims to create a deep learning model for coronary heart disease prediction, with the help of feature selection techniques and hyper-parameter optimization for model tuning. The paper's objective is not only to improve the accuracy but also to experimentally prove that the unique combination of feature selection technique and deep neural network model used by us is very effective in making predictions with low false negatives and can be practically used in hospitals. The dataset used in our research is BFRSS-2015 which is a collection of survey data coordinated by centers of Disease Control and Prevention (CDC). It contains basic health-related and daily routine information, which can help individuals to diagnose the disease in the very early stage.

3. Methodology

This section describes the techniques adopted in the paper, with proper definitions and equations, along with the dataset description of the dataset used and pre-processing techniques employed.

Table 1
The state-of-art representation of previous work.

Ref.	Year	Dataset	Description	Results
[52]	2018	Cancer dataset and two different Heart Disease datasets	Modified Bagging method for classification of imbalanced dataset	Accuracy: 78.15%
[15]	2019	UCI Heart Disease dataset	Ensemble of machine learning algorithms on Rapid Miner	Accuracy: 93.19% with DT
[16]	2021	UCI Heart disease dataset	ML models such as LR, KNN, SVM, RF, and DT and Deep Neural Nets with feature selection and outlier detection	Accuracy: 94.20% using deep neural net
[17]	2021	UCI Heart disease dataset	Hybrid DT, RF and KNN with Bagging and AdaBoost techniques With Relief and LASSO Feature Selection Techniques (RFBM)	Accuracy: 94.20% using RFBM and Relief feature selection methods
[21]	2022	Sathvi dataset	ML classifier such as NB, XGB, SVM, kNN, CatBoost, and MLP algorithms with feature selection and cross-validation	Accuracy: 98.11% using CatBoost
[22]	2022	Cleveland heart dataset	ML algorithms such as kNN, DT, NB, SVM, RF, and LR classifiers	Accuracy: 92.30% with LR
[23]	2022	UCI Heart disease dataset	ML algorithm based classifiers; AB, LR, ET, MNB, CART, SVM, LDA, RF, and XGB	Accuracy: 96.72% with SVM
[24]	2022	UCI Heart disease dataset	ML algorithm such as kNN, DT, RF, LR, and SVM	Accuracy: 82.18% using RF
[53]	2022	UCI Heart Disease dataset	SVM, kNN, Decision Trees, Naive Bayes, and Random Forest classifiers are used	Accuracy: 88.35% with RF
[54]	2023	UCI Heart Disease dataset	Data normalization followed by stacking of ensemble classifiers and cross-validation	Accuracy: 91.80%
[29]	2023	BRFSS 2015 Heart Disease dataset	ML algorithm such as XGB, bagging, RF, DT, knn, and NB based classification	Accuracy: 91.30% using XgBoost Model
[28]	2022	BRFSS 2015 Heart Disease dataset	ML algorithm such as XGB, LightGBM, RF, NB, DT, kNN, and AdaBoost based classification with sampling technique for class imbalance	Accuracy: 97.10% using Random forest

FNR: False Negative Rate, FPR: False Positive Rate, PRNU: Photo Response non-uniformity, CNN: convolution neural networks, RNN: Recurrent Neural Network, LSTM: Long-sort term memory, SCN: Spatiotemporal Convolutional Networks.

Table 2
Data classes description.

Classes	Number of person's samples
Heart diseases present	23,893
Heart diseases not present	229,787

3.1. Dataset description

One of the largest surveys carried out in the USA is the Behavioural Risk Factor Surveillance System (BRFSS). Each year, BRFSS conducted over 400,000 adult telephone interviews. The questions focus on potential threats to health, long-term illnesses, and the utilization of preventative care. All states of U.S.A., the District of Columbia, and three other U.S. territories are included in this poll.

This analysis relies on information from the 2015 BRFSS report. Heart disease risk variables were surveyed in the original data (441,455 responses, 330 characteristics). There are a total of 253,680 cases in the cleaned dataset, along with 21 quantitative features. No values are missing from this data collection. A binary class consisting of 229,787 responds from individuals who were not diagnosed with heart disease and the remaining 23,893 responses from those who were diagnosed with heart disease is created. The samples of this dataset are given in Table 2 below. The following are the final outputs of the dataset. High blood pressure, high cholesterol, smoking, race, obesity, sex, diabetes, diet, exercise, age, alcohol consumption, body mass index, family income, marital status, education, health insurance, checkup frequency, and mental health are all part of the feature set.

3.2. Data balancing using cluster the abundant class

In a "imbalanced" dataset, the distribution of the classes that are produced is skewed. When referring to output classes, "majority classes" refer to those with a greater number of samples, while "minority classes" refer to those with less. One of the significant issues in the BRFSS 2015 Heart Disease dataset is class imbalance [55]. A classification dataset which has bias in the target class proportions is called an imbalanced dataset. To remove the imbalance in the dataset the paper uses clustering of the abundant classes. The major challenge in BRFSS 2015 heart disease dataset is Unbalanced training happens when one class is represented by a large number of samples and another class is represented by a small number of samples. Let represent an unbalanced data collection, with $_{min}$ and $_{maj}$ representing the subsets of samples from the majority and minority classes, respectively. The explanation of the datasets balancing ratio is [56]:

$$X = \frac{|\ _{min}|}{|\ _{maj}|} \tag{1}$$

where || denotes the cardinality of a set.

The balancing process is analogous to resample X into a new dataset X_{rest} such that $X > X_{rest}$, clustering the abundant class in $_{maj}$ groups, with $_{maj}$ being the number of cases in $_{maj}$. Only the medoid (cluster centre) is retained for each group. The model is then trained using only the rarest class and the medoids [52].

3.3. Co-relation based feature selection (CFS)

The correlation between two random variables is a well-known indicator of similarity. The correlation coefficient between two random variables if they are linearly dependent on one another is ±1. The correlation coefficient is 0 if there is no correlation between the variables. The correlation coefficient is not affected by translation or scaling. The average absolute mutual correlation of a feature's total features is computed by CFS after it evaluates all mutual correlations for all feature pairs. For subsequent training and prediction purposes, the feature with the highest mutual correlation and which directly impact the target value will be chosen.

3.4. Randomized Search CV optimization (RSCV)

A "fit" and a "score" approach is implemented by Randomized Search CV optimization to fine-tune the model's parameters. The model parameters of the estimator are optimized by a cross-validated search across a range of parameter choices. As opposed to GridSearchCV, where all possible parameter values are tested, this method just tries a small subset of them from the chosen distributions. The total number

of parameter settings that are tested is provided by N_iter. The first step is to define the factor we intend to take into consideration, and then determine the top criteria for this selection. Now, RandomizedSearchCV (RSCV) will be implemented and data will be fitted. Finally, it will result in the optimal parameter values for the proposed model.

3.5. LSTM

The LSTM is an enhanced version of the recurrent neural network model (RNN) [57]. It is designed to address the vanishing gradient problem and capture long-term dependencies in sequential data. It introduces a memory cell, which allows it to store and retrieve information over extended sequences. LSTMs use a gating mechanism to control the flow of information, allowing them to mitigate the vanishing gradient problem that can hinder the training of traditional RNNs. This makes them more effective at learning and retaining information over longer sequences. It has four components; memory cell, input gate, output gate, and forget gate. LSTMs are capable of learning and maintaining information over long sequences, making them well-suited for tasks where understanding context is crucial, are certain advantages of LSTM.

3.6. BiLSTM

BiLSTM flips information by superimposing an LSTM layer [57]. After that, the two LSTM layers' outputs are merged using average, sum, multiplication, and concatenation. The underlying argument for utilizing BiLSTM is that every component of an input sequence comprises data from both the past and the present [58]. BiLSTM produces more accurate output due to incorporating LSTM layers in both directions. BiLSTM appears complex, yet because of a thorough awareness of the environment, the outcomes are positive. Bi-directional context and enhanced contextual understanding are advantages of using BiLSTM over LSTM [37]. In this research, the BRFSS data is subjected to a BiLSTM model to predict heart disease among patients [59].

3.7. GRU

GRU is a simplified variant of RNN with improved performance [60,61]. GRU, like LSTM. GRU, also called another variant of the LSTM architecture [57], is designed to simplify the gating mechanism and reduce the number of parameters, which makes them computationally more efficient and easier to train. They merge the input and forget gates of the LSTM into a single "update gate" and also combine the memory cell state and hidden state, resulting in a more streamlined architecture compared to LSTM. The key components of GRU are; the update gate, reset gate, and candidate activation. GRUs tend to perform well even with limited training data due to their simpler architecture and reduced number of parameters, and also require less computation compared to LSTM, these are the few advantages of using GRU over LSTM [59].

4. Proposed model

An explanation of the proposed model for predicting cardiovascular diseases is provided here, including how the requisite data is prepared, what features are extracted and chosen, and how the resulting predictions are classified. Fig. 1 shows the proposed model flowchart with the parameters used.

Fig. 1 provide a detailed work flow of the proposed model in prediction heart disease among patients. For experiment purpose, the paper has utilized BRFSS 2015 heart disease dataset, which contains the clean data of 253,680 patients, the detailed dataset description is provided in Section 3.1. In order to balance the unbalanced BRFSS heart disease dataset, cluster abundant class technique is employed. The balanced BRFSS heart disease data is now utilized for extracting the best correlation between the features using pearson correlation based feature selection (CFS) technique. For subsequent training and prediction purposes, the feature with the highest mutual correlation and which directly impact the target value. Later the dataset is divided into of training and testing for classifying the patients with heart disease and without any heart disease. This is done with the combination of BiLSTM-GRU model, a hybrid RNN model hyper-parameter tuned using randomized search cross-validation optimization (RSCV) technique. The detailed description of the work flow and working of each step is elaborated in the below subsections. Algorithm 1 gives the description of the steps followed in the proposed model.

Algorithm 1 Proposed algorithm for CHD classification

1: **Results:** Performance of model will observed by confusion matrix parameter, ROC curve an Convergence curve
2: **Input:** Dataset of BRFSS 2015 heart disease.
3: **Output:** Detection of Coronary Heart Disease;
4: **Procedure:**
5: **Step 1:** Data Balancing using cluster the abundant class
6: **Step 2:** The features are selected using RSCV optimization model:
7: **Step 3:** Use sequential BiLSTM-GRU, a deep learning model with 100 epochs, for the classification of Coronary Heart Disease
8: **Step 4:** Generate Confusion metric and store the result in terms of accuracy, precision (aka PPV), recall (aka sensitivity), f1-score, specificity and NPV value.
9: **Step 5:** Repeat Step 2 and Step 4 for the models for comparison with the proposed model;
10: **End**

4.1. Balanced data and feature selection

In this proposed method data class is balanced by the cluster abundant data class method. Cluster formation is based on the centre of the cluster (medoid). The balance data contain equal data in labelled classes. Balanced classes are now used for feature selection. An efficient well-known technique namely co-relation score between input and output feature mapping. Fig. 2 shows the co-relation heatmap between input and selected features based on a high co-relation score. The relationship between continuous and class features may be identified via a correlational approach with correlation score greater than 0.2 in BRFSS balanced dataset. Most of the features of dataset which affect heart diseases are hypertension, cholesterol, BMI, smoking, diabetes, exercise, stroke, food, and heavy alcohol consumption, which has greater correlation score.

4.2. Dataset splitting

After successful data balancing and feature selection and before classification, it is necessary to split the data into training and testing. In this research, we have divided the dataset into multiple training-testing ratios that are 60:40, 65:35, 70:30, 75:25, and 80:20, the training and testing ratio of 70:30 respectively gives the best results. That means, 70% BRFSS heart disease data employed for training our proposed model and remaining 30% data is utilized for testing the accuracy of the suggested model. In order to test the robustness of the proposed model, 10-fold cross-validation is applied after classification. The dataset is partitioned into 10 equal-sized folds or subsets, and the process is automated using functions that begin by randomly shuffling the dataset to ensure that the samples are distributed randomly among the folds. This step is necessary to avoid bias in the data ordering.

4.3. Hyper-parameter tuning and classification

Model parameters play the most important role in classification. In this proposed model, randomized search CV (RSCV) optimization is used for hyperparameter tuning. RSCV parameters like estimator =

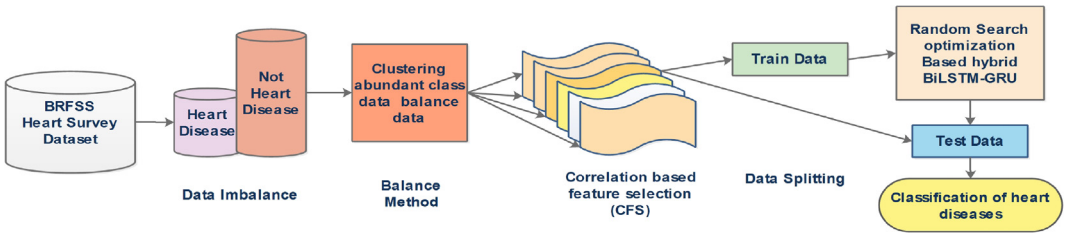

Fig. 1. Proposed work flowchart.

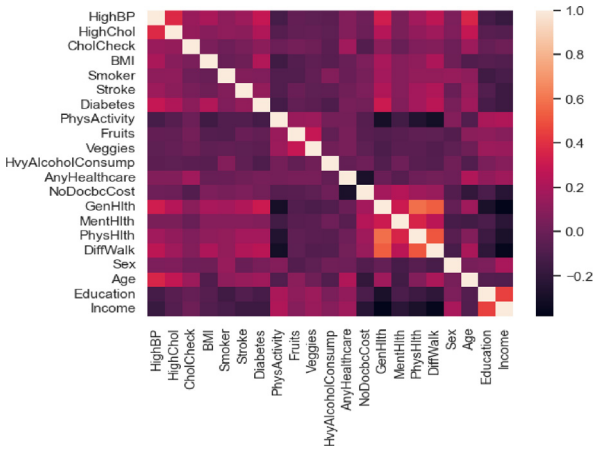

Fig. 2. Co-relation heatmap.

Table 3
Sequential model layers parameter.

Layer1	Input layer(Training data)
Layer2	BiLSTM(filter = 64)
Layer3	GRU(filter = 16)
Layer4	Dropout(0.4)
Layer5	Dense(Filter = 1, activation = sigmoid)
Compiler	Loss(Binary crossentropy), Optimizer(Adam)
Fitting	Epoch(100), Batch-size(20)

BiLSTM and GRU estimator for score calculation, model parameters for hyperparameters are the number of filter = [64, 32, 16, 8], the number of iteration = 100, cross-validation = 10 and random-state = 25 are used for the optimized filter in the model. After the fitted RSCV model optimizes the filter size for the proposed model at 64 for BiLSTM and at 16 for GRU. Fig. 3 shows the proposed model flow with working and parameters used. One of BiLSTM's key strengths is that it incorporates historical and current data into its models. GRU is quicker and requires less memory than LSTM, however, LSTM is more accurate when working with datasets that contain longer sequences. Table 3 shows the Sequential BiLSTM-GRU Model Layers Parameter details. In the proposed model layer 1 take input as training data and passed it to the next layer2 (BilSTM) with a filter size value 64 and the output of this layer is passed as input for layer3 (GRU) with a filter size value of 16. The dropout (0.4) layer 4 is used for reduced model over-fitting. The Dense layer5 is used with one filter size and activation function sigmoid. This model complied with Loss set as binary cross-entropy and Optimizer set as Adam. This model trains with `epochs` = 100 and `batchsize` = 20. Finally, this model outperforms as compared to other similar deep-learning models.

5. Result

This section presents the results and discussion of the proposed model and other similar Deep Learning models based on model conversion curve, receiver operating characteristic (ROC) curve, cross-validation techniques, and comparative analysis. The proposed model was created with the Python programming language and the Deep Learning Tensor-Flow powerful library.

Table 4
Confusion matrix parameter calculation.

Parameter	Formula
Positive predicated value(PPV)	$\frac{TP}{TP+FP} * 100$
Recall or Sensitivity	$\frac{TP}{TP+FN} * 100$
Specificity	$\frac{TN}{FP+TN} * 100$
Negative predicted Value(NPV)	$\frac{TN}{FN+TN} * 100$
F1-Score	$\frac{2*(PPV*Sensitivity)}{PPV+Sensitivity} * 100$
Accuracy	$\frac{TP+TN}{TP+FN+FN+TN} * 100$

Table 5
Proposed and other model comparison.

Models	Accuracy	SEN	SPE	PPV	NPV	F1-Score
GRU	96.80	98.40	95.20	95.23	98.51	96.82
LSTM	96.87	98.40	95.41	95.30	96.81	96.87
BiLSTM-LSTM	97.50	98.50	95.55	95.54	98.53	97.05
BiLSTM-GRU	97.74	98.58	95.82	95.78	98.59	97.17
Proposed Model	98.28	98.86	97.57	97.57	97.56	98.21

NPV, Negative Predicted Values: PPV, Positive Predictive Value; SEN, Sensitivity; SPE, Specificity; All values shown in percentages.

5.1. Metrics based performance

In this work, we considered heart disease prediction from the BRFSS-2015 dataset. The data is classified into two classes; Heart disease present and not present. Classification of binary task evaluated based on the confusion matrix parameters; Sensitivity (SEN), Specificity (SPE), Negative Predicted Value (NPV), Positive predicted value (PPV), F1-Score, Accuracy. Table 4 specifies the formulation of the below-mentioned evaluation parameters. These parameters play a crucial role in assessing the effectiveness of classification models. These metrics provide insights into the model's performance by quantifying the true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions. A thorough assessment employing a mix of these metrics aids researchers and practitioners in making knowledgeable judgments regarding the capabilities and limitations of their models. The proposed model confusion metrics parameter are compared with GRU, LSTM, BiLSTM-LSTM, and BiLSTM-GRU models.

A comparison analysis of standard techniques with our proposed model is shown below in Table 5. Better results of our proposed model can be seen in this table. As an example, the proposed model Accuracy (98.28%) is better than GRU (96.80%), LSTM (96.87%), BiLSTM-LSTM (97.50%), BiLSTM (97.74%) which indicates the correct values are predicted. Similarly, the F1-Score (98.21%) value is also better than other models. Fig. 4 shows the metric Parameters Graphical Comparison of models.

Fig. 3. Proposed model working flowchart.

Fig. 4. Metric parameters graphical comparison of models.

5.2. Receiver operating characteristic (ROC)

The receiver operating characteristic (ROC) curve is used to determine whether or not heart disease exists and to validate the performance measure using the confusion metric. The ROC curve is used to assess the number of true positives and false positives in test data. The true positive rate on the Y-axis and the false positive rate on the X-axis has been plotted together with the ROC curve. The ROC curve for the proposed model and compared with other models is demonstrated in Fig. 5. The area under the ROC curve (AUC) is a two-dimensional measurement of this relationship, which depicts the proposed model outperformed other models.

5.3. Convergence curve

While training and validating a model, a convergence curve reveals the optimal value of the learning parameter in terms of the model's accuracy relative to the loss function. Fig. 6 illustrates the Convergence curves of the proposed and other models. The subfigures 6(a) GRU, 6(b) LSTM, 6(c) BiLSTM-LSTM, 6(d) BiLSTM-GRU and, 6(e) Proposed models are shown the 100-epoch tanning and validation accuracy curves. Both the training and validation accuracy of the proposed model indicates a faster convergence rate and an accuracy of 98.28%, which is better than the accuracy of the other created models.

This research work focuses on a classification task. During the model training process, the Binary Cross-Entropy loss, also referred to as logarithmic loss or log loss, is utilized for calculating the loss in each epoch. A lower loss value accompanied by higher accuracy indicates the model's excellence. The validation loss values for various models are as follows: GRU (0.3293), LSTM (0.3130), BiLSTM-LSTM (0.2531), BiLSTM-GRU (0.2261), and the Proposed Model (0.1721). It is noteworthy that the proposed model demonstrates a lower loss compared to the other models, suggesting its superior performance.

Fig. 5. ROC comparison.

5.4. Cross-validation

In order to acquire quantitative statistical results, predictive models are first constructed on a dataset, and then resampling processes are used to test the models. In conclusion, methods of cross-validation are put to use in order to assess the resiliency of our model. The suggested model was trained and tested using the usual 10-fold cross-validation procedure, which produced an average classification accuracy of 98.05%.

As the model's accuracy is reported as 98.28%, and the average 10-fold cross-validation accuracy is approximately 98.05%. These reported values are accompanied by error margins of about ±0.009%. These error margins establish a range within which the true accuracy values are expected to lie with a 95% confidence level. It is crucial to take these error margins into account when interpreting and comparing accuracy results. They offer valuable insights into the uncertainty associated with the model's performance estimates, enhancing the accuracy assessment's reliability and context.

6. Discussion

The paper proposed an RSCV optimization-based hybrid deep neural network learning model for predicting coronary heart diseases (CHD).